

Final

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9:30

TOPIC NAME: Network layer: IP addressing

DAY: _____

TIME: _____

DATE: 12/11/25

Network layer logical addresses are 32 bits

IP header → source and destination header

IP address → 32 bit

↳ divided into 4 part (each part 8 bit)

1st 3 → network part

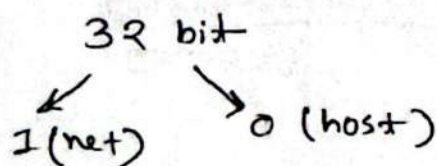
last 1 → host part

Network part same means same network & belong to it.

network part / host part কতটুকু, সেটা ফিক্স না
সিদ্ধান্ত ২৪ bit → network bit

$$\text{host bit} = 32 - 24 = 8$$

Subnet mask → কতগুলো network part



prefix mask of /24 or subnet 255.255.255.0

IP address: 10.24.36.2/4

host 28, network 4

$$\begin{array}{c} \text{network} \\ \overline{1111}0000 \cdot 00000000 \cdot 00000000 \cdot 00000000 \\ \downarrow \\ 240 \cdot 0 \cdot 0 \cdot 0 \end{array}$$

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TOPIC NAME: _____

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Subnet mask: 255.255.255.192

$\frac{1}{128}$ $\frac{1}{64}$ $\frac{0}{32}$ $\frac{0}{16}$ $\frac{0}{8}$ $\frac{0}{4}$ $\frac{0}{2}$ $\frac{0}{1}$ } Shortcut binary

→ 192

26 → prefix mask

IP address AND subnet mask = network part
as it is

and host part 0

↓
network address

Broadcast address: host → 1

→ indicate করে প্রত্যেকটি device কে

host: 00000000 → network } device কে দিও
 11111111 → broadcast } পারবে না

$$2^8 = 256$$

$$256 - 2 = 254$$

2 host bit - 2 = usable IP address

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Unicast: single pc to packet forwarding

Broadcast: limited,

limited: own network as a separate network
destination fix

255.255.255.255

↳ not a subnet mask

limited broadcast as packet router forward
not a

Multicast: 1 or group of device to forwarding

IP address: 224-239

Loopback: device as a separate - for practice

192.168.10.127/25 → broadcast

127 → $\begin{array}{c} 01111111 \\ \hline \text{N} \quad \text{H} \end{array}$

192.168.11.0/23 → Host address

11 → $\begin{array}{c} 00001011 \\ \hline \text{N} \quad \text{H} \end{array}$

192.168.10.128/25 → network

128 → $\begin{array}{c} 10000000 \\ \hline \text{N} \quad \text{H} \end{array}$

192.168.0.255/16 → host

00000000 11111111

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TOPIC NAME _____

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* 192.168.165.127 / 20 → usable IP addresses
host addresses

$$165 \rightarrow \frac{10100101}{n \quad h}$$

Subnet mask, network address, broadcast, total
usable
and unusable

$$SM \rightarrow 11111111 \ 11111111 \ 11110000 \ 00000000$$

$$255.255.240.0$$

$$NA \rightarrow 192.168.160. \frac{0}{h} \rightarrow \text{all } 0$$

$$\hookrightarrow N/A \rightarrow \frac{10100000}{n \quad h} \rightarrow 160$$

$$\hookrightarrow N/A \cap h = 0$$

$$BA \rightarrow 192.168.175. \frac{255}{h} \rightarrow \text{all } 1$$

$$\hookrightarrow B/A \rightarrow \frac{10101111}{n \quad h} \rightarrow 175$$

$$\hookrightarrow B/A \cap h = 1$$

$$192.168.160.1 \rightarrow \text{1st usable}$$

$$192.168.160.2 \rightarrow \text{2nd usable}$$

$$192.168.175.255 \rightarrow \text{last usable}$$

$$192.168.175.253 \rightarrow \text{2nd last usable}$$

$$\text{total usable} = 2^{12} - 2 = 4094$$

TOPIC NAME: Lecture 7

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Network Jagen: IP Addressing

IP Address: 192.168.10.1

↳ 32 bit 10 1 ডাটা ডাটা ক্রি

$$32/4 = 8 \text{ bit}$$

↳ octet

1st 3 part → network

last part → host

Network/host part কয়েকটা fixed না।

Router এর ২য় interface (২য়) থাকে, ততসূত্রে network

২য়।

1st

Network এর undetachable কিছু device থাকলে, তাদের network part same ২য়।

Subnet mask: কয়েকটা network part

↳ 32 bit : network (1), host (0)

prefix mark: /24

Subnet mask: 255.255.255.0

} same

* IP address: 10.24.36.2/12, what's the Subnet mask?

11111111.11110000.00000000.00000000

↓
255↓
240↓
0↓
0

255.240.0.0

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* IP address : 10.24.34.2
subnet mask: 255.255.224.0
what's the prefix mask?

11111111.11111111.11100000.00000000
prefix mask: 19

*

IP address : 135.15.2.1

10000111 00001111 00000010 00000001

Subnet mask: 255.255.0.0

11111111 11111111 00000000 00000000

Network address : 135.15.0.0
↳ 1 এর সাথে যা AND করে, 0 এর সাথে যা AND করে, 0 হৈ আসবে

10000111 00001111 00000000 00000000

↳

IP এর part

↳

subnet এর 0

IP Address AND subnet mask = network part as it is and host part 0
↓
network address

Network address : কোনো device কে indicate করে না

host bit = 0

Broadcast address : network এর সব address কে indicate করে।

host bit = 1

একটা network এর সবাইকে মাঝে মাঝে broadcast address use করা। Particular device কে address

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বর্তমান খারচা না।

Host address:

$$2^8 = 256$$

$$\text{usable} = 2^8 - 2 = 256 - 2 = 254$$

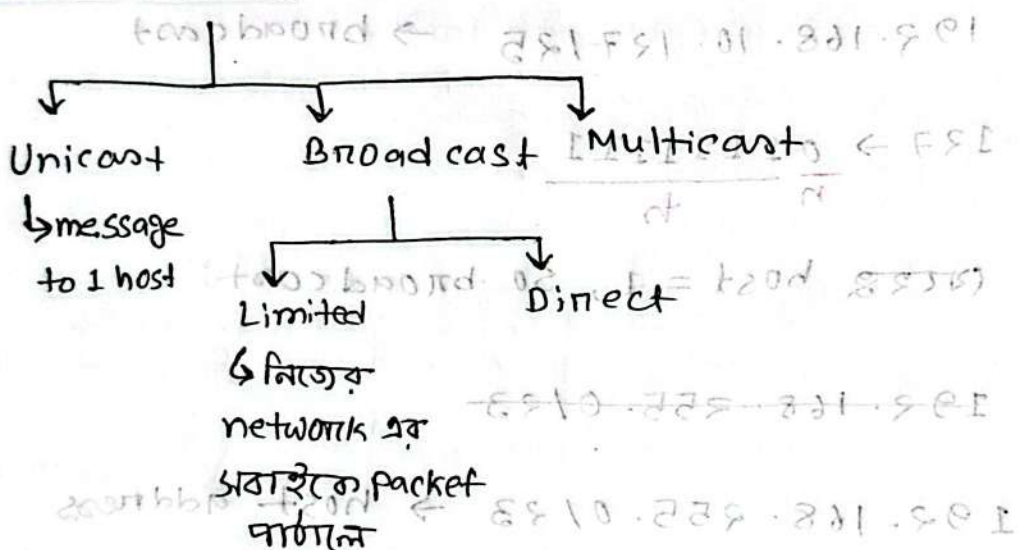
00000000 $\rightarrow$ network

11111111 $\rightarrow$ broadcast

এই ২ টা বাকি আর usable address

$$\text{Usable IP address} = 2^{\text{host bit}} - 2$$

Types of Casting



Limited broadcast ৩ Destination IP: 255.255.255.255

*** Limited broadcast এর packet router software ব্যবহার করে না

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Directed broadcast: নিজের network ব্যতীত অন্য network এর সাথেই যোগাযোগ

Multicast: group of device to যোগাযোগ

Loopback address: নিজের সাথে communicate করে। device এর ব্যতীত অন্য না।

Practice

192.168.10.127/25 → broadcast

127 → $\frac{01111111}{n}$ $\frac{11111111}{h}$

(যেহেতু host = 1, so broadcast)

~~192.168.255.0/23~~

192.168.255.0/23 → host addresses

255.0

$\frac{11111111}{n}$ $\frac{00000000}{h}$

(যেহেতু host bit is 1, 0 উভয়ই অন্য → host addresses)

TOPIC NAME: _____

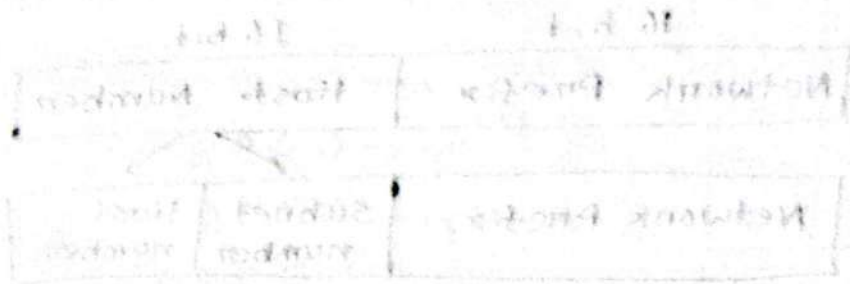
DAY: _____

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192.168.10.128 / 25 → network address

128 → $\frac{10000000}{n}$

192.168.0.255 / 16 → host address



TOPIC NAME: Subnetting

DAY: _____

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Problem: Lack of IP address

Solutions:

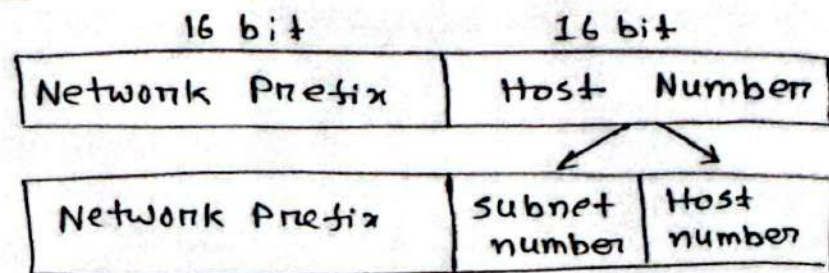
Long term:

- i) change to IP version 6 $\rightarrow$ 128 bit
- ii) Plenty of addresses using a different scheme

Short term:

- i) Use subnetting to avoid wasting addresses
- ii) Use private addresses locally and NAT for internet access - lets many host share a few public addresses.

Subnetting



If subnet 2 bit $\rightarrow 2^2 = 4$ combination
Host 14 bit

If subnet 3 bit $\rightarrow 2^3 = 8$ combination
Host 13 bit

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TOPIC NAME: _____

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Two methods of subnetting:

- i) Fixed length Subnet Masking $\rightarrow$ prefix fixed/same $\hookrightarrow$ equal
- ii) Variable Length Masking $\rightarrow$ equity

Fixed length Subnetting

Subnetting જૂથમાં network address ફિઝ વગર અડી.

step:

- i) count network bits $\leftarrow$ 8 $\leftarrow$ 4 bit + 200

Network 4 ટી નાસ્ટલ શાબલ network address નાસ્ટલ

4 ટી 1 So combination નાસ્ટલ 4 ટી 1 2 ટી bit $\rightarrow 2^2 = 4$

- ii) decide how many bits we need

0 0 | 0 0 0 0 0 0 $\rightarrow 0$

0 1 | 0 0 0 0 0 0 $\rightarrow 64$

1 0 | 0 0 0 0 0 0 $\rightarrow 128$

1 1 | 0 0 0 0 0 0 $\rightarrow 192$

new network bit increased by 2

Olden: 200.20.20.0/24

host bit = $32 - 24 = 8$

$$24 + 2 = 26$$

updated: 200.20.20.0/26

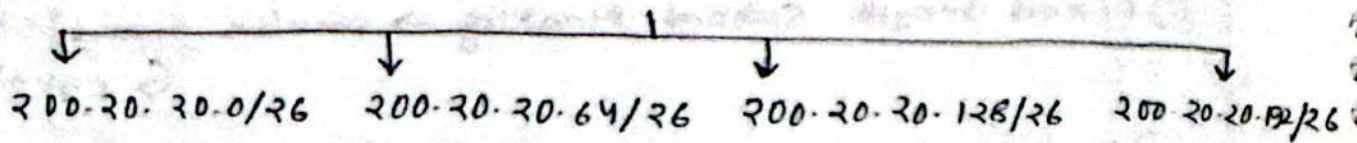
host bit = $32 - 26 = 6$ usable IP address = $2^6 - 2 = 62$
We need 2 IP address. Waste = $62 - 2 = 60$

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200.20.20.0/24



* If there is 7 network

Then we need 7 network address.

So in this case, we need 3 bits.

192.168.80.0/24

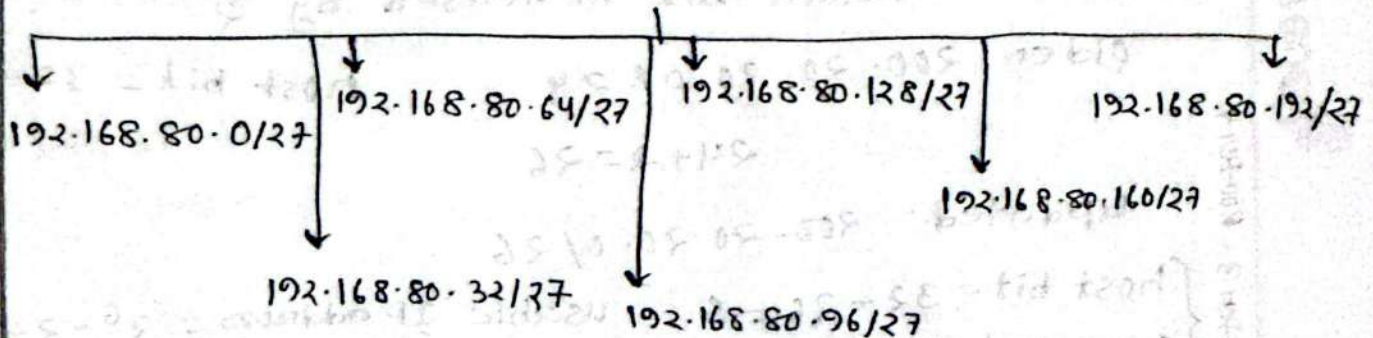
host bit $\rightarrow 8$

$$24 + 3 = 27$$

0 0 0	0 0 0 0 0	$\rightarrow 0$
0 0 1	0 0 0 0 0	$\rightarrow 32$
0 1 0	0 0 0 0 0	$\rightarrow 64$
0 1 1	0 0 0 0 0	$\rightarrow 96$
1 0 0	0 0 0 0 0	$\rightarrow 128$
1 0 1	0 0 0 0 0	$\rightarrow 160$
1 1 0	0 0 0 0 0	$\rightarrow 192$
1 1 1	0 0 0 0 0	$\rightarrow 224$

7 network address

192.168.80.0/24



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Fixed length Subnet masking $\rightarrow$ Lots of waste

Solution: Variable length Subnet masking (VLSM)

Variable length Subnet Masking

i) 200 hosts. Assign a block size 256 IPs

$$2^7 = 128$$

but we need 200

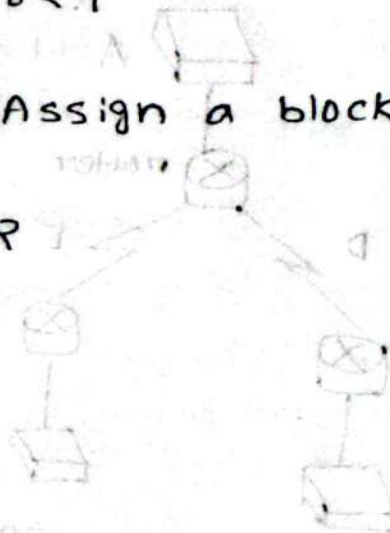
$$\text{So } 2^8 = 256$$

ii) 1000 hosts. Assign a block size 1024 IPs

$$2^{10} = 1024$$

iii) 2 hosts. Assign a block size 2 IPs.

$$2^1 = 2$$



TOPIC NAME: VLSM example 1

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* $192.168.16.0/24$

5 network

$192.168.16.0/24$

host bit = $32 - 24 = 8$

FLS 97 twhm:

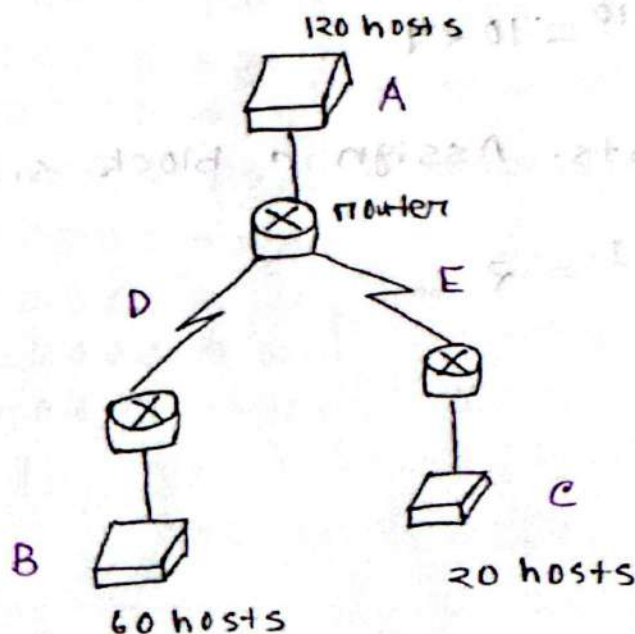
network 5 tr, so bit 3. host = $8 - 3 = 5$

$2^5 - 2 = 30 \rightarrow$ usable

Given 120 host

usable 30

so we can't solve it by FLS.



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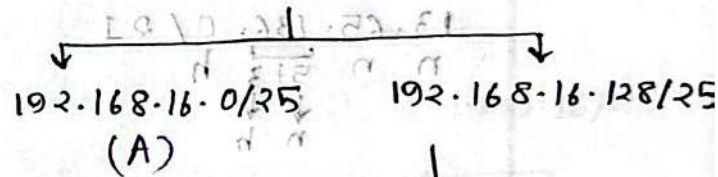
TIME : _____

DATE : / /

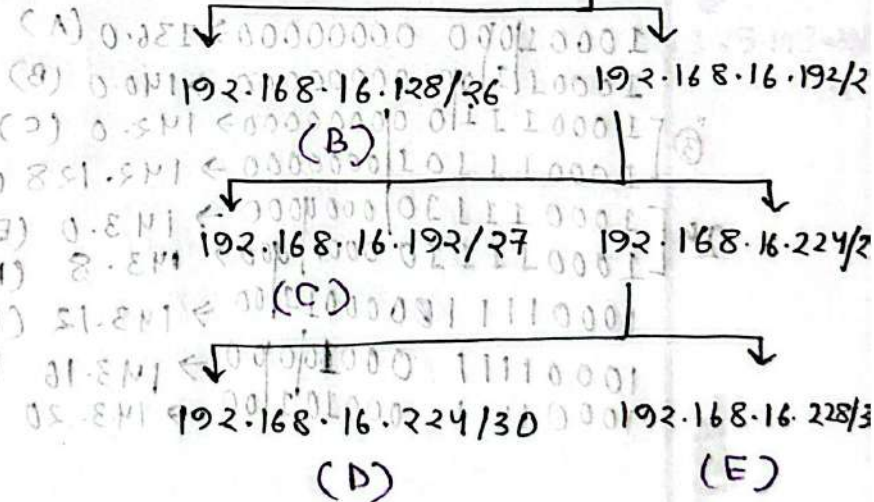
Name	Host	Host + 2	Block size	Host bits	Prefix	Waste
A	120	122	128	7 → 20	25 → 32-7	128-122=6
B	60	62	64	6	26	2
C	20	22	32	5	27	10
D	2	4	4	2	30	0
E	2	4	4	2	30	0

Total waste = 18

192.168.16.0/24



0|0 0 0 0 0 0 0 0 → 0 (A)
 1|0 0 0 0 0 0 0 0 → 128 (B)
 1 1|0 0 0 0 0 0 0 0 → 192 (C)
 1 1 1|0 0 0 0 0 0 0 0 → 224 (D)
 1 1 1 0 0|0 0 0 0 0 0 0 0 → 228 (E)



network 4 (2 digit) માટે

અગાળા change રાવના ।

TOPIC NAME: VLSM Example 2

DAY: _____

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Name	Host	Address/ Host + 2	Block size	Host bits	Prefix
A	950	952	1024	10	22
B	500	512	512	9	23
C	120	122	128	7	25
D	91	93	128	7	25
E	3	5	8	3	29
F	2	4	4	2	30
G	2	4	4	2	30
H	2	4	4	2	30

13.65.136.0/21
 n n $\frac{5+3}{\downarrow}$ h
 n h

10001000 00000000 → 136.0 (A)

10001100 00000000 → 140.0 (B)

③ 10001110 00000000 → 142.0 (C)

10001110 10000000 → 142.128 (D)

20 10001110 00000000 → 143.0 (E)

10001110 00010000 → 143.8 (F)

10001111 00000100 → 143.12 (G)

10001111 00010000 → 143.16 (H)

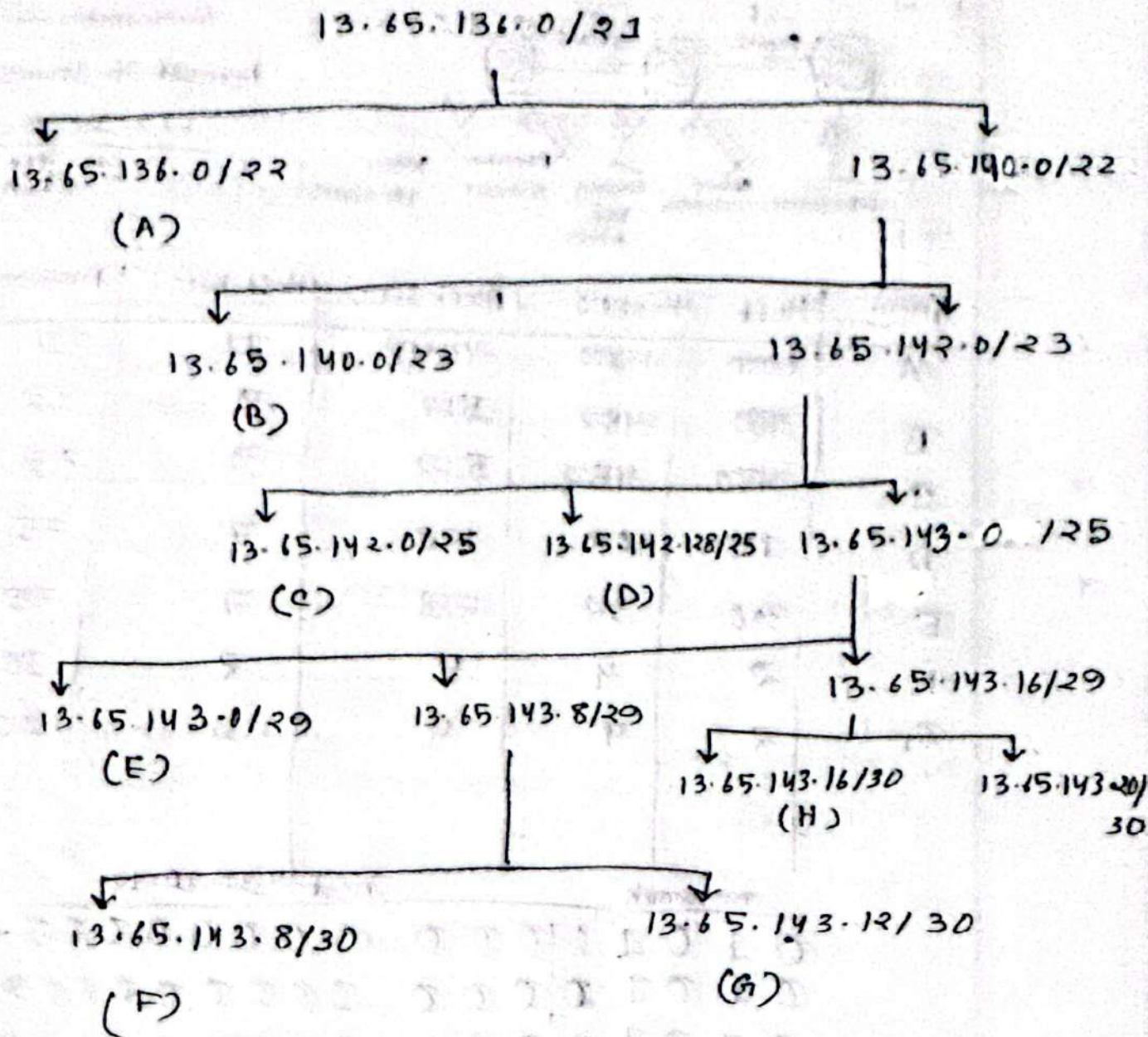
10001111 00010100 → 143.20

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TOPIC NAME: _____

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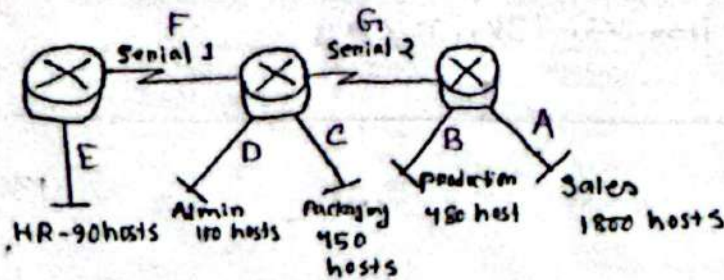


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TOPIC NAME: VLSM Example

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network = 7

highest to lowest

$$\frac{173}{n} \cdot \frac{32}{n} \cdot \frac{80}{414} \cdot \frac{0}{h} / 20$$

Name	Host	Host + 2	Block size	Host bits	Prefix
A	1800	1802	2048	11	21
B	480	482	512	9	23
C	450	452	512	9	23
D	100	102	128	7	25
E	90	92	128	7	25
F	2	4	4	2	30
G	2	4	4	2	30

network	host → 32-20=12
0 1 0 1 0	0 0 0 0 0 0 0 0 0 0 0 0 → 80.0/20
0 1 0 1 0	0 0 0 0 0 0 0 0 0 0 0 0 → 80.0/21 (A)
0 1 0 1 1	0 0 0 0 0 0 0 0 0 0 0 0 → 88.0/21
0 1 0 1 1 0 0	0 0 0 0 0 0 0 0 0 0 0 0 → 88.0/23 (B)
0 1 0 1 1 0 1	0 0 0 0 0 0 0 0 0 0 0 0 → 90.0/23 (C)
0 1 0 1 1 1 0	0 0 0 0 0 0 0 0 0 0 0 0 → 92.0/23
0 1 0 1 1 1 1	0 0 0 0 0 0 0 0 0 0 0 0 → 94.0/23
0 1 0 1 1 1 0 0	0 0 0 0 0 0 0 0 0 0 0 0 → 92.0/25 (D)
0 1 0 1 1 1 0 0	1 0 0 0 0 0 0 0 0 0 0 0 → 92.128/25 (E)
0 1 0 1 1 1 0 1	0 0 0 0 0 0 0 0 0 0 0 0 → 93.0/25
0 1 0 1 1 1 0 1	0 0 0 0 0 0 0 0 0 0 0 0 → 93.0/30 (F)
0 1 0 1 1 1 0 1	0 0 0 0 0 1 0 0 0 0 0 0 → 93.4/30 (G)

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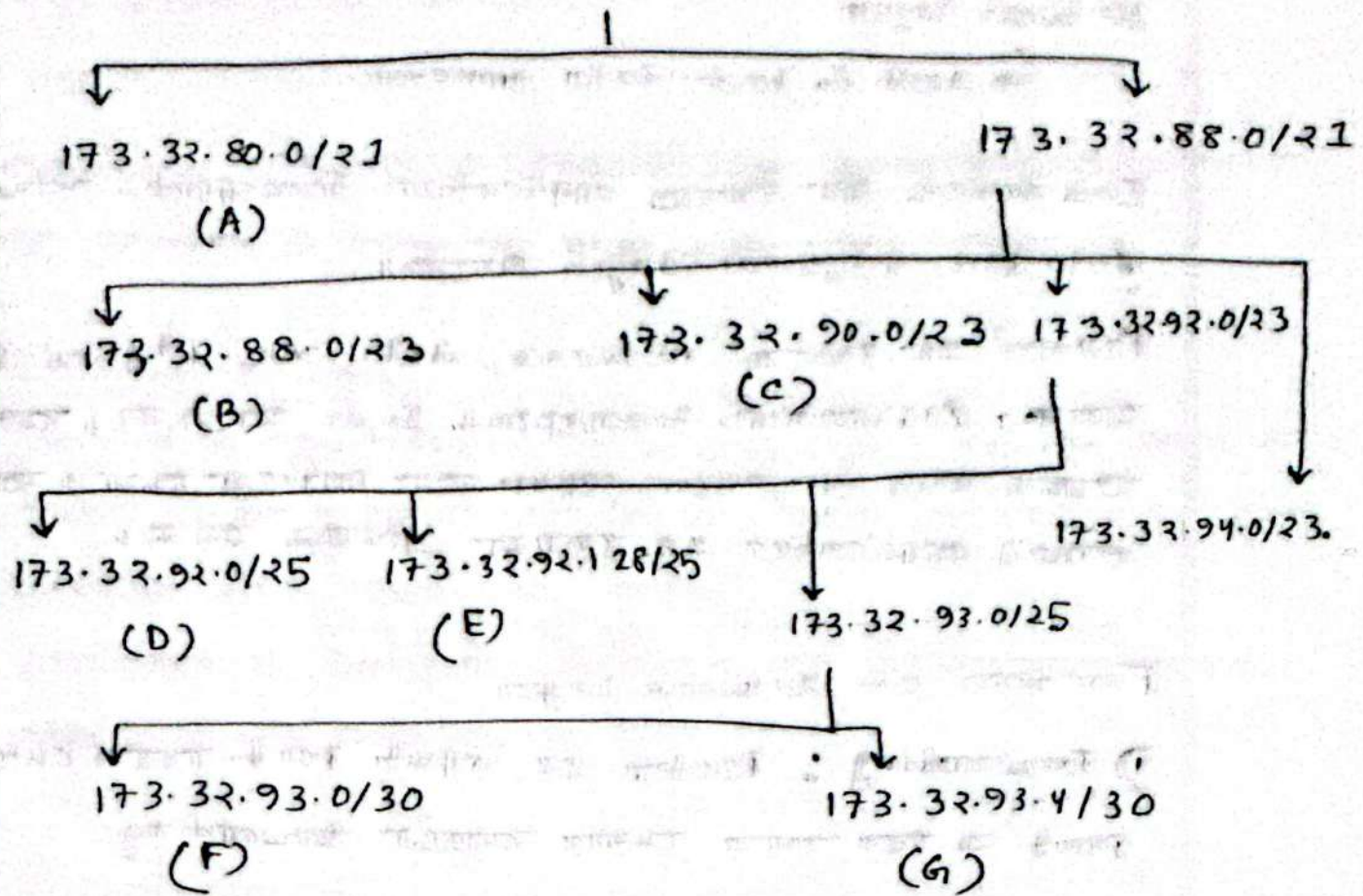
TOPIC NAME: _____

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173.32.80.0/20



TOPIC NAME: Lecture 9

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Network Layer - IPv4 Functions

Network layer

↳ host to host data transfer

End device এর ক্ষেত্রে application, transport, network, data link, physical layer থাকে।

Router এর ক্ষেত্রে network, data link, physical layer থাকে। Application, transport layer থাকে না। কারণ তাদের কাছে যা আসে, তারা তা পাঠিয়ে দেয়। তাদের email application এর কোনো প্রয়োজন হয় না।

Functions of Network Layer

- i) Forwarding : Router এর input port থেকে output port এ বের করে দেয়ার কাজটা forwarding.
- ii) Routing: Determine করে 1 টা packet source থেকে destination এ কীভাবে যাবে

Routing table create করে routing এর মাধ্যমে। Table দিয়ে forwarding এর মাধ্যমে 1 টা port থেকে আরেকটা port এ পাঠিয়ে দেবে।

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Connection and Connection-less Service

TCP $\Rightarrow$ connection-oriented

UDP $\Rightarrow$ connection less

Datagram network $\Rightarrow$ network layer connectionless service

VC network $\Rightarrow$ network layer connection service

Datagram Network

No end to end connection $\Rightarrow$ fixed path create করে না।
অর্থাৎ 1 path দিয়ে সব packet পাঠায় না। যখন যে path available থাকে, তখন তমটা পাঠায়।

Internet ও Datagram Network এর মধ্যে কাজ করে।
অর্থাৎ fixed path create করে না।

Virtual circuit

fixed path create করে।

VPN: Privacy দরকার হলে fixed path লাগে। যেহেতু
virtual circuit high cost. So low cost alternative হচ্ছে VPN.
VPN virtual circuit এর concept follow করে। VPN একটি
dedicated / fixed path create করে। তাদের নিজস্ব router
নেই। যে internet এর মধ্যে যে router আছে, তা use করে
fixed path create করে।

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* Is I use TCP in transport layer and datagram in network layer, will that be creating problem?

TCP $\rightarrow$ connection oriented

Datagram $\rightarrow$ connection less

No, then that's not a problem.

process to process connection আর নাহি তা check করে transport layer. connection থাকলে কোন দিক দিয়ে তা pass করতে হবে তাই decide করে network layer. So network and transport layer এর কাজ different. So No problem.

Functions of Router

Forwarding এর কাজ করে Router এর Switching Fabric নামে একটি part. আর তা

Routing processor $\rightarrow$ routing এর কাজ করে

Queuing $\rightarrow$ process করতে time লাগে

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TOPIC NAME: IPv4 Functions

DAY :

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IP Fragmentation & Reassembly

যুটী PC directly connected না থাকলে, যখন এ টা PC অন্য PC কে কিছু পাঠাতে চায়, তখন Link এর মাধ্যমে পাঠায়।

প্রত্যেকটি link এর bandwidth different. যদি Link MTU (max transfer size) 1500 bytes হয়, তাহলে link highest 1500 bytes একসাথে পাঠাতে পারে। Suppose, PC 4000 bytes পাঠাতে চায়, কিন্তু MTU 1500 bytes. At this moment, Packet কে ভাগ করতে হবে। এর জন্য fragmentation প্রয়োজন।

Packet / Datagram $\rightarrow$ Data + header

Fragment করতে total packet-র fragment করার প্রয়োজন
নেই, শুধু- data-র fragment করতে হয়।

* Suppose total length = 5140, data = 5140 - 20 = 5120 bytes
 ↓
 header

IP fragments:

Total length = 1500, data = $1500 - 20 = 1480$

$$5120 - 1480 = 3640$$

$$3640 - 1480 = 2160$$

$$2160 - 1480 = 680$$

$$680 + 20 = 700$$

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Identifien: Identifien টাখ বুঝাও-গাছি তা কোন original packet এর অন্তর্ভুক্ত।

Fragment offset = initial byte number $\div$ 8

Data Bytes	Fragment offset
0 - 1479	$0/8 = 0$
1480 - 2959	$1480/8 = 185$
2960 - 4439	$2960/8 = 370$
4440 - 5119	$4440/8 = 555$

* why we divided it by 8 ?

- Datagram format এ fragment offset 13 bit allocate

করা। কিন্তু byte number usually 16 bit এর হয়।

So 3 bit right shift করলে, $2 \times 2 \times 2 = 8$ হয়।

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ICMP $\rightarrow$ Internet control message protocol

This is a helping protocol of Network layer.

i) PING $\rightarrow$ ২ টা device এর মাধ্যমে connectivity check

ii) Traceroute $\rightarrow$ একটি device থেকে অন্য device এ packet

পাঠানো রুটগুলো hop দিয়ে দাবি করে হয়, তার address
লাভ দাবি।

PING: PING করে device এ যা identify করে IP address দিয়ে,

কিন্তু URL দিলে কি তা কাজ করে? $\rightarrow$ yes, কারণ URL থেকে

DNS তা IP address এ convert করে দেয়।

PING এর ক্ষেত্রে আমরা packet ১ বার কেন sent করি?

Suppose, connection মাঝে কিছু সমস্যা অনুভব হলে।

ক্ষেত্রে ১ টা পাঠানো connection break হলে response

আমাদের connection possible না। তাই আমরা ১ বার

sent করি। এতে ১ বার connection হলে at least

১ টা packet তার পৌঁছাবে। তাই ১ বার sent করি।

আবার suppose, packet ২ টার response আসলে, বা কি

২ টার আসলে না তাহলে ক্ষেত্রে 50% packet loss.

১ টার response ই যদি আসে তাহলে 0% packet loss.

অর্থাৎ connection ভালো।

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ICMP message

Memorize

Type	Code	Description
0	0	echo reply (ping)
3	3	dest port unreachable
3	4	Fragmentation needed and DF flag set
8	0	Echo request
11	0	Time to live exceeded in transit

Ping Attack

ICMP PING flood attack / DOS Attack:



Denial of service

Attacker source address বার বার change করবে এবং destination যাক target এবং continuously PING দাখাবে। Server PING handle করতে গিয়ে crash করবে, then real customer to service provide করতে পারবে না। এটাই DOS Attack বলে।

ICMP DDOS attack - Zombie Attack:



Distributed Denial of Service

Attacker অনেক device to mail পাঠাবে, যেখানে attachment

GOOD LUCK

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এ virus/Trojan দিয়ে দিবে। কোড/attachment click করলে virus install হয়ে যাবে। Then virus target server কে ping করতে থাকবে। server ping reply দিতে দিচ্ছে না, then actual customer কে service দিতে পারবে না।

ICMP DDoS attack - Packet magnification (or ICMP smurf):

Source	Destination
Victim's IP	Broadcast of victim network

Attacker ping করলে সব device ping এর reply দিবে victim কে, অর্থাৎ source কে। তখন যে তার buffer এ store করতে করতে buffer full হয়ে যাবে, তখন victim PC hang হয়ে যাবে।

Traceroute

Tool used to trace path from source to destination host

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DHCPv4 & NAT

DHCP → Dynamic Host Configuration Protocol

যাখন automatically network & host connect

হয়, তখন host-কে IP address দিয়ে হয়।

Why DHCP?

DHCP network এর device সূত্র to automatically IP address দিয়ে হয়, এর সাথে DNS, Subnet mask, Default gateway সব provide করে। Manually provide করা feasible or scalable না। তাই DHCP use করা হয়।

DHCP Operations

DHCP দুই ভাবে করা যায়। (i) Router, (ii) Dedicated server

The DHCPv4 server dynamically assigns or leases an IPv4 address from a pool of addresses for a limited period of time chosen by the server or until the client no longer needs the address.

DHCPv4 works in a client/server mode.

DHCP client port → 68

DHCP server port → 67

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Steps to obtain a Lease (DORA)

i) DHCP Discover (Broadcast)

Finds the DHCP server(s) on the network.

Client doesn't know where its IP addresses are, so it sent request to everyone, so it indicates Broadcast. In this case, only DHCP server will reply, others will drop the packet.

ii) DHCP Offer (Unicast)

Contains an available IP address to lease.

iii) DHCP request (Broadcast)

It sends an acceptance notice to the selected server and an implicit decline to any other servers.

So, here it informs which IP it takes and which it doesn't take → Broadcast.

Also used for lease renewal and verification.

iv) DHCP acknowledgment (Unicast)

Verifies the lease information.

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Steps to Renew a Lease

i) DHCP Request.

Suppose, server has a pool of IP provide করে।

এক্ষেত্রে server দ্বারা হওয়া যায় client server

কো lease extend করতে পারে।

ii) DHCP acknowledgement

Client যখন Lease extend করতে চায় তখন

server positive acknowledge দ্বারা এবং lease extend করে।

এক্ষেত্রে, client request করার ক্ষেত্রে যদি (delay

হয়- এবং server দ্বারা দ্রুত মাধ্যমে তখন server

IP pool এ back করে এবং অন্য client-কে IP (same)

provide করে। তখন previous client request

করলে server NACK (negative acknowledgment)

provide করে। তখন client আবার DORA র

1 টা step follow করে।

GOOD LUCK

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DHCP Relay

Router won't forward Broadcast Packet (255.255.255.255).
So DHCP Relay is used to transfer the packet.

Steps to configure a CISCO IOS DHCP Server

Step 1: Exclude IPV4 addresses.

A single address or a range of addresses can be excluded by specifying the low-address and high-address of the range. Excluded addresses should be those addresses that are assigned to routers, servers, printers and other devices that have been, or will be, manually configured.

ip dhcp excluded-address [low-address] [high-address]

Step 2: Define a DHCPV4 pool name

ip dhcp pool pool-name

Step 3: Configure the DHCPV4 pool.

GOOD LUCK

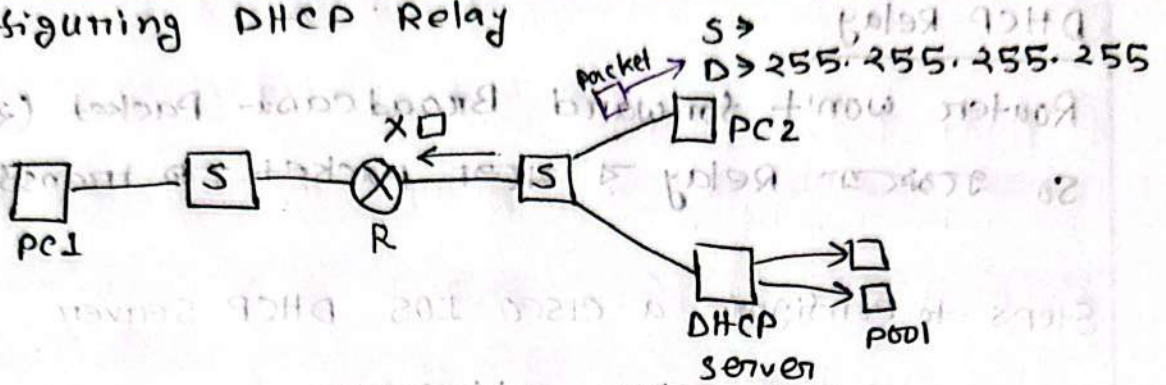
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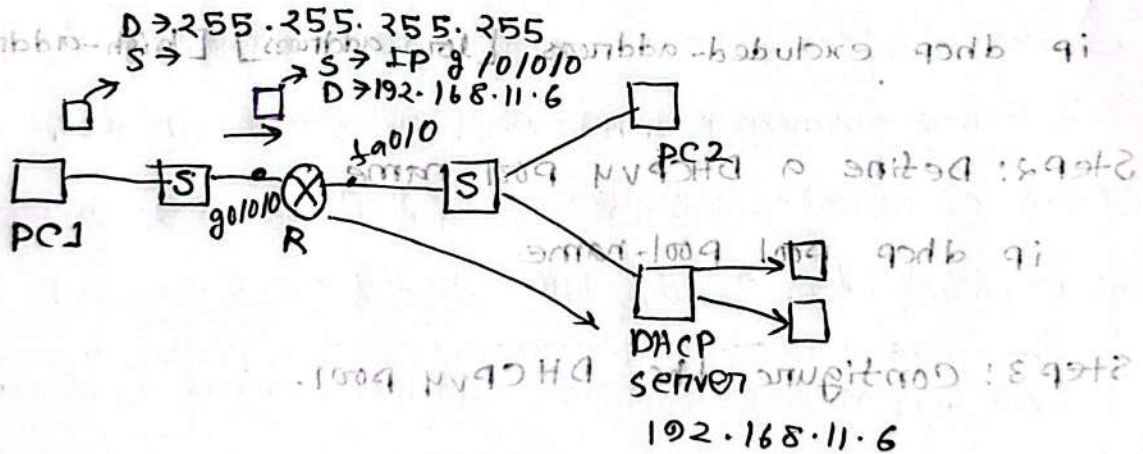
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Configuring DHCP Relay



PC2 থেকে packet switch এ forward করার পর
 এ যখন router এ যায়, তখন Router তা drop
 করে দেয়। এরপর server এর কাছে জানে যে
 DHCP packet নিয়ে identify করে, সেই তার
 network এর। তখন DHCP server pool থেকে
 address নিয়ে PC2 তে দিয়ে দেয়।



Interface দিয়ে request আসবে, তাই interface
 এ চুকানো। তার ip helper address set করানো,
 তারপর থেকে যখন Router DHCP এর কোনো packet
 দেখবে 80/0/0 থেকে, তখন তা unicast করে DHCP
 server এ পাঠিয়ে দিবে।

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Packet Relay করার পর S → IP 0/0/0, D → 192.168.11.6
 DHCP server এ request করার পর check করলে
 0 255.255.255.255 না। অর্থাৎ এটি তার নিজের network
 এর নয়। 0/0/0 দিয়ে identify করে address কোন network
 এ আছে। যদি identify করে 2 no network এর। অর্থাৎ
 2 no pool থেকে address নিয়ে পাঠিয়ে দিলে।
 S → server, D → 0/0/0. অর্থাৎ Request Router এ চলে
 আসে, then রুট PC1 কে forward করে দিলে।
 তারপর PC1 তার configuration করতে পারে।

NAT (Network Address Translation)

Problem: Lack of IP address

Short term solution: NAT

NAT uses 2 type of address

i) Private ii) Public

Private Address

Class	Activity Type	Activity Name
A	10.0.0.0 - 10.255.255.255	10.0.0.0/8
B	172.16.0.0 - 172.31.255.255	172.16.0.0/12
C	192.168.0.0 - 192.168.255.255	192.168.0.0/16

Private → can't access Internet

Public → can access Internet

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Public Internet Addresses:

- i) These are for public routing
- ii) can be advertised globally

What is NAT?

Private Address $\longleftrightarrow$ Public Address
NAT

Network need a mechanism to translate private addresses to public addresses at the edge of their network that works in both directions.

How NAT works:

When a host on the inside (private) network wants to access a host on the outside (public) network, the packet is sent to the border gateway/router.

Inside network: Usually an organization's LAN

Outside network: Usually the Internet but it can be

any network. The address of the destination device.

Local address: appears inside portion of network $\rightarrow$ private

Global address: appears on the outside portion of network $\rightarrow$ public

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Example: NAT: Sending

Source address: 192.168.10.10 (private)

Destination address: 209.165.201.1

Packet border gateway router is a router that router that source address is private address. Private address from internet is not possible. So, in this case, we need to convert the private address to public address. Then the 'Inside local address' will be stored in NAT-Table and it places the converted public address in source address. Then the server will get the packet and will reply.

Example: NAT: Receiving

Source address: 209.165.201.1

Destination address: 209.165.200.226 (public)

Server will send the packet to Packet border gateway router. But the destination address is a public address so it'll not enter in LAN. So we need to give PC's address and will fetch it from NAT Table.

Now DA: 192.168.10.10

So PC1 will receive the packet and then will process it. After this, we will delete the entry from NAT Table, so that will give back the address to NAT Pool.

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PAT (NAT overload)

Map multiple private IPv4 addresses to a single public IPv4 address on a few addresses.

PC with same port first request to router
then port to change to next available
port to provide unique connection.

DORA

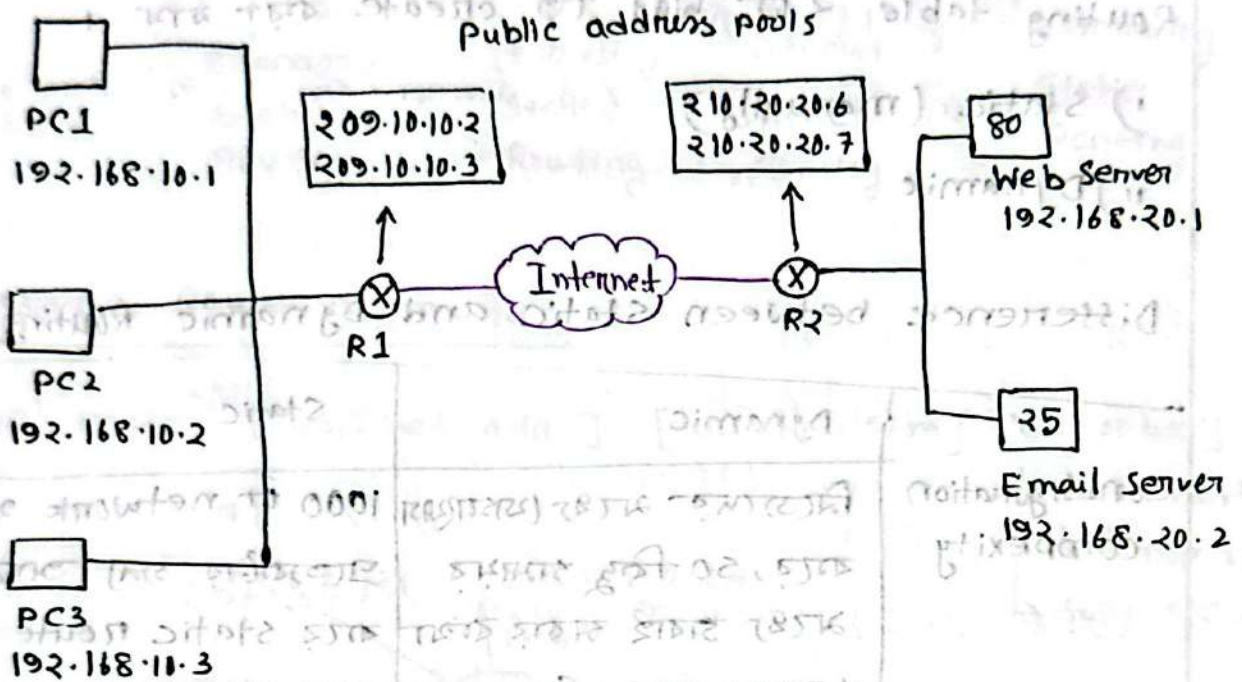
	S.IP	D.IP	S.MAC	D.MAC	D.Port	S.Port
Discover (Broadcast)	0.0.0.0	255.255.255.255	Host A	FFFF:FFFF:FFFF	67	68
Offer (Unicast)	172.16.4.250	172.16.4.1	DHCP server	Host A	68	67
Request (Broadcast)	172.16.4.1	255.255.255.255	Host A	FFFFFFFFFFFF	67	68
Ack (Unicast)	172.16.4.250	172.16.4.1	DHCP server	Host A	68	67

TOPIC NAME: NAT & PAT Simulation

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PC1 (ii)

Source Address	Destination Address	D/A	S/A
192.168.10.1:1444	210.20.20.6:80	209.10.10.2:1444	192.168.20.1:80
209.10.10.2:1444	192.168.20.1:80	192.168.10.1:1444	210.20.20.6:80

R1'S NAT Table

Inside local	Inside Global
192.168.10.1:1444	209.10.10.2:1444
192.168.10.2:1445	209.10.10.2:1445
192.168.10.2:1444	209.10.10.2:1446

R2'S NAT Table

Inside Local	Inside Global
192.168.20.1:80	210.20.20.6:80
192.168.20.2:25	210.20.20.6:25

PC2 (i)

S/A	D/A	S/D/A	D/A
192.168.10.2:1445	210.20.20.6:25	192.168.20.2:25	209.10.10.2:1445
209.10.10.2:1445	192.168.20.2:25	210.20.20.6:25	192.168.10.2:1445

PC3 (iii)

S/A	D/A	D/A S/A	D/A
192.168.10.3:1444	210.20.20.6:80	192.168.20.1:80	209.10.10.2:1446
209.10.10.2:1446	192.168.20.1:80	210.20.20.6:80	192.168.10.2:1444

GOOD LUCK

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IPv4 Static Routing

Routing table 2 টা way to create করা যায়।

i) static (manually)

ii) Dynamic

Difference between Static and Dynamic Routing:

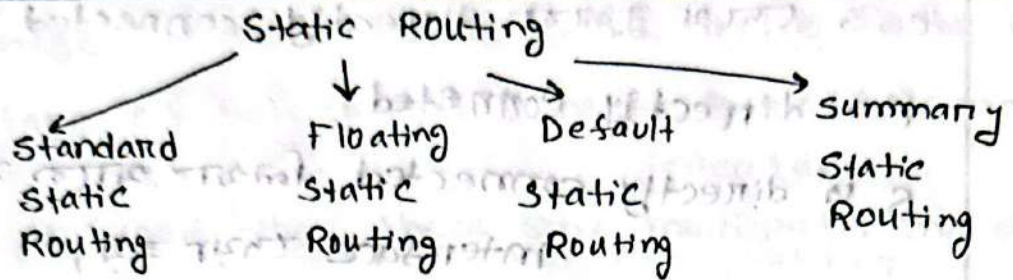
	Dynamic	Static
Configuration complexity	নিজস্বের মাধ্যমে (যাচাযাচ) করে, 50 কিছু সমস্যার মাধ্যমে করা হয়। So, generally independent of the network size and less configuration complexity.	1000 টি network থাকলে প্রত্যেকটির জন্য আলাদা করে static route create করা লাগবে। (ii) অর্থাৎ size বাড়লে configuration complexity বাড়বে।
Topology changes	নিজস্বের মাধ্যমে communication করে জানতে পারে। So automatically adapts to topology changes.	Topology এ change হলে প্রত্যেকটিতে আলাদা করে জানানো লাগবে। here administrator intervention required
Scaling	Suitable for simple and complex topologies	Suitable for simple topologies
Security	less secure	more secure
Resource usage	uses CPU, memory, link bandwidth	no extra resources needed
Predictability	Route depends on the current topology	Route to destination is always the same

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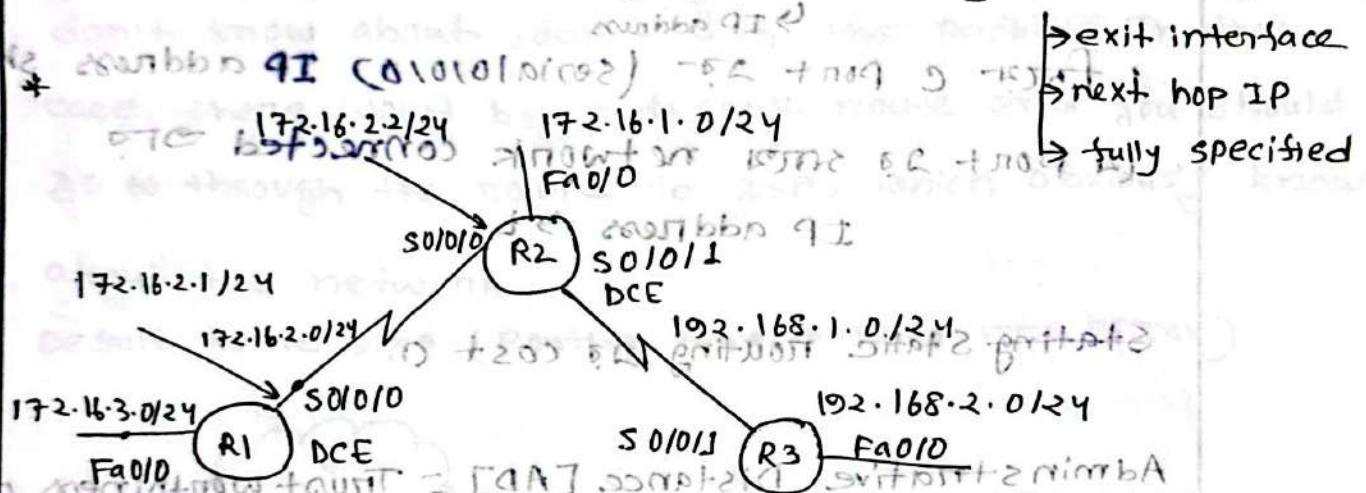
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Static Route Command:

`ip route [dest. net. add.] [dest. net. s/m] [exit option] [AD]`



R1 is static route create for 172.16.1.0/24 network

`ip route 172.16.1.0 255.255.255.0 S0/0/0` → exit interface

`172.16.2.2` → next hop IP

`ip route 172.16.1.0 255.255.255.0 S0/0/0 172.16.2.2` → fully specified

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C & S router directly connected router

C → directly connected

S is directly connected with S router exit interface that is,

C 172.16.2.0/24 is directly connected, Serial 0/0/0

L 172.16.2.1/32 [172.16.2.0/24] Serial 0/0/0

↳ IP address

1. First C port is (Serial 0/0/0) IP address show that

(2) port is same network connected that

IP address →

Setting static routing that cost 0

Administrative Distance [AD] = Trustworthiness of the source of the route

AD of C = 0

AD of S = 1

AD and cost are lower that better.

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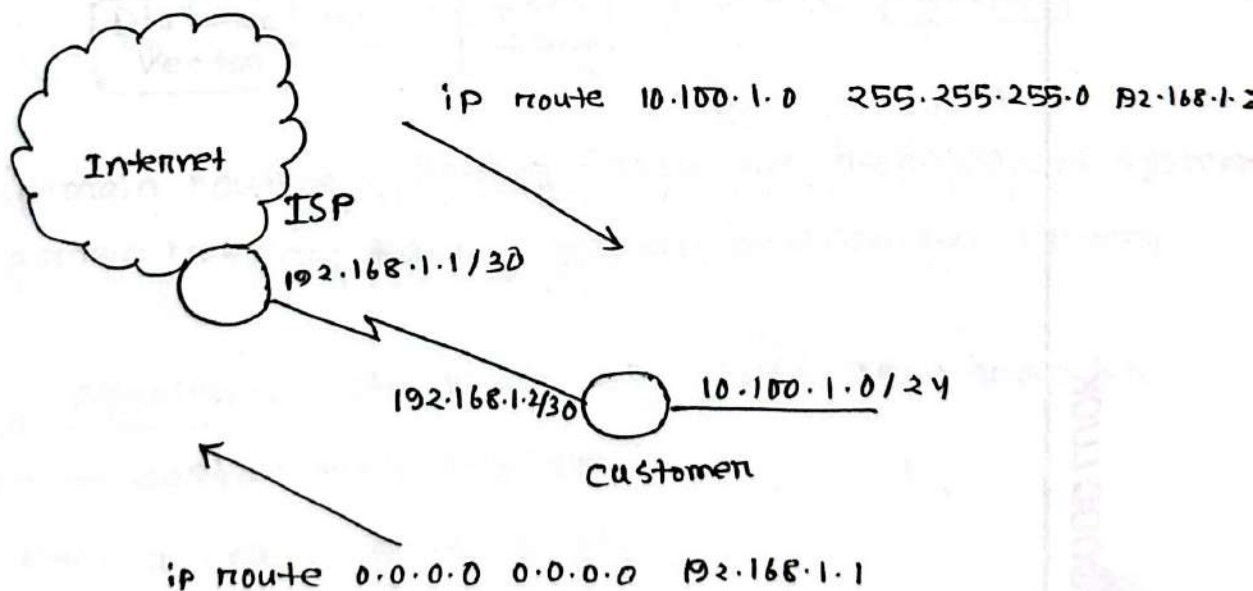
Disadvantage of Next Hop IP: Recursive Route Lookup

Disadvantage of fully specified static route: multi-access interface.

It get confused when there are multi path and drop packet for this.

Default Static Route: If there are packets that you don't know about, don't drop the packet. In that case, there would be a default route and you should go through the route to ISP, which obviously know about the network.

Default Route $\rightarrow S^*$ (Routing table 6 সবসর নতুন থাকবে)



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Floating static Route: primary route down হয়

তাল backup route দিয়া হয়।

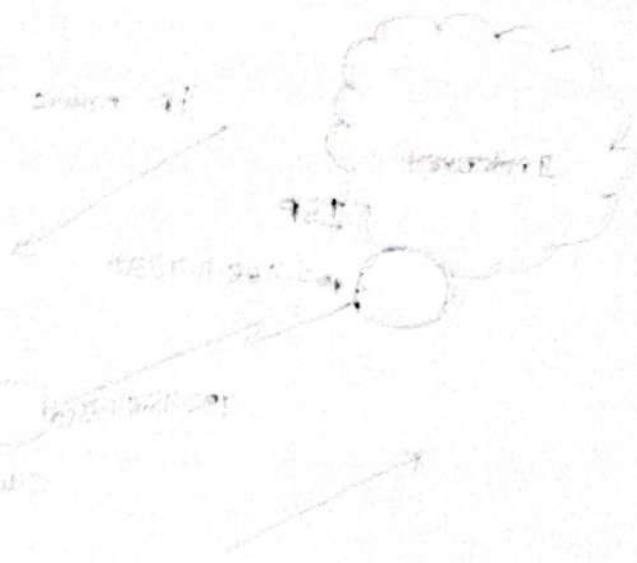
backup route ও AD এর value তালি থাকে, যত router configured না হয়। AD এর value যত

কম তত ভাল। Primary route এ $AD = 1$ । তখন

Route primary route choose করে। Somehow

primary route down হলে backup route use

করে।



GOOD LUCK

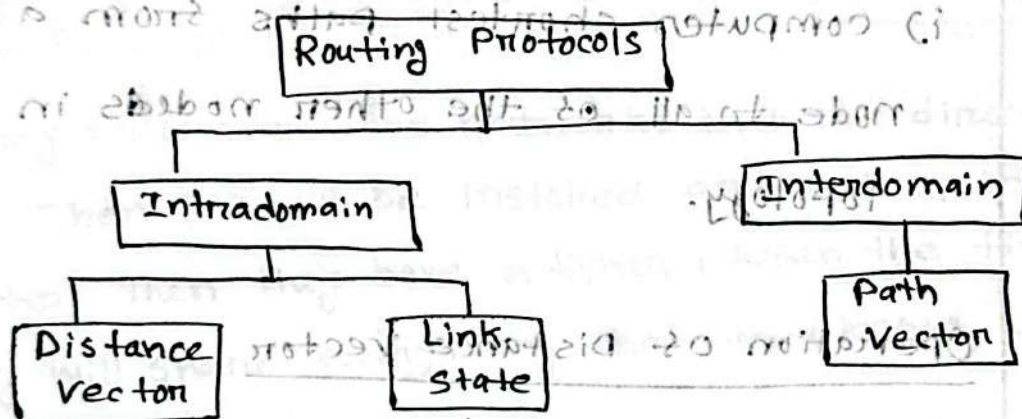
Distance Vector Routing

Dynamic Routing: no configuration and must know

→ Distance vector

→ Link state

Autonomous System: Internet is divided into autonomous systems. This is a group of networks and routers under the authority of a single administration.



Intra-domain routing: Routing inside an autonomous system

Inter-domain routing: Routing between autonomous system

Routing Algorithm: It finds a "good" path from the source to destination router.

Good Path: Least cost path

Routing Algorithm →

→ Global

→ Decentralized

TOPIC NAME: _____

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Global: Gather the information of full topology and run the algorithm on it.

Decentralized: After getting partial information of the topology, run the algorithm on it.

Distance Vector Algorithm

Based on Bellman-Ford Algorithm.

- i) computes shortest paths from a single source node to all of the other nodes in a weighted topology.

Operation of Distance Vector

Periodic Updates:

- i) Periodically broadcast the entire routing table to each of its neighbors. (RIP - every 30 seconds)

↳ inefficient.

- ii) Router is only aware of the:

- i) network addresses of its own interfaces

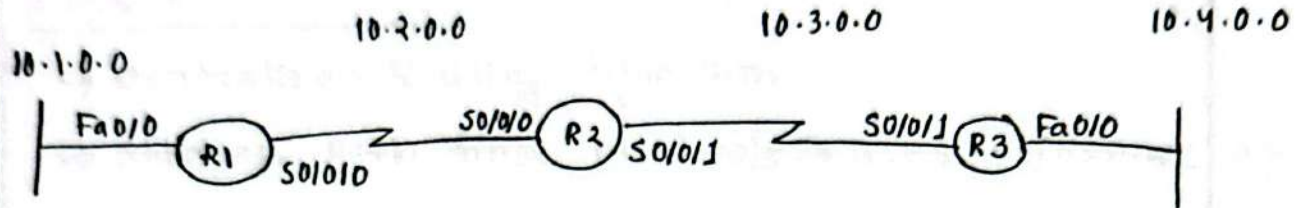
- ii) Network addresses the neighbors running the same routing protocol.

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Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	1
10.4.0.0	S0/0/1	2

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/1	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0
10.2.0.0	S0/0/1	1
10.1.0.0	S0/0/1	2

Initially, they know about the networks that are directly connected. Then RIP will be installed and set up in every router. Then they have a timer, when the time is over, they will share everything that they know with other neighbour.

R2 will share

10.2.0.0 is already exist in R1. It will update if the cost is less than its own.

R1 don't have the info of 10.3.0.0, so add 10.3.0.0.

Interface will be S0/0/0, the point where neighbour will enter. From R1 to R2, there is 1 hop and then R2 to 10.3.0.0, it's 0. So $1 + 0 = 1 = \text{cost}$.

R3 don't have the info of 10.2.0.0, so add it. $\text{cost} = 1 + 0 = 1$

R1 will share

10.1.0.0 added, interface = the point from where R2 will leave

Link state Routing

↳ centralized Routing Algorithm

↳ Shortest Path first protocols → uses Dijkstra's algorithm

Link states don't use hop count.

This algorithm accumulates costs along each path, from source to destination.

Fast convergence

Link state Routing Process → 5 steps

Step 1: Directly connected networks

Step 2: Hello Packets

Hello packet will reply when there are routing device. LAN will not reply.

If 3 hello packets won't reply, we consider that the connection with neighbour could be down.

↳ keep alive function

Step 3: Build the Link-state Packet (LSP)

directly connected packet's information will be stored in Link-state packet.

Step 4: Flooding Link-state Packets

Only sent to neighbors.

Step 5: Constructing a Link-state Database

TOPIC NAME: IPv6

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DATE: / /

Internet Protocol, Version 6 (IPv6)

IPv6 $\rightarrow$ 128 bit

Sixed length, 40 byte header

No fragmentation allowed

Traffic class $\rightarrow$ QoS provide

Payload length $\rightarrow$ the amount of data

Payload length + header (40) = packet size

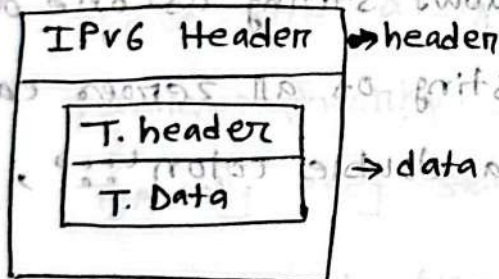
Transport Layer

T. Header	T. Data
-----------	---------

 $\rightarrow$ Network Layer Data

TCP header value = 6

Network Layer

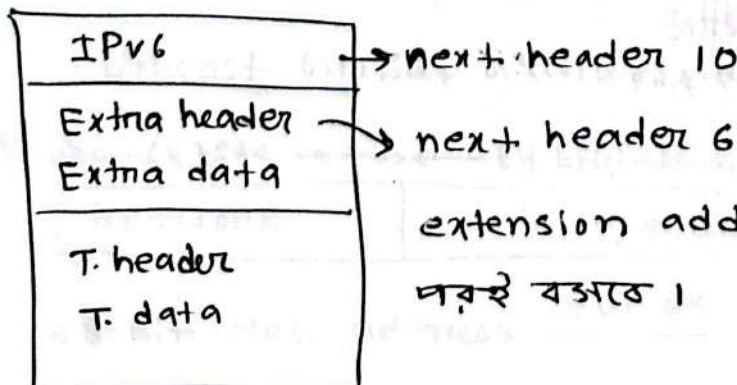


next header = 6

extension $\rightarrow$

extra header	extra data
--------------	------------

$\rightarrow$ suppose 10



extension add হাল তা ফিচ IPv6 এ
পরই বসবে।

GOOD LUCK

TOPIC NAME: _____

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128 bit IPv6 → divided into eight 16 bits blocks → binary
 → converted into hex and separated by ':'
 → called string notation

IPv6 Representation

Rule 1: The leading zeroes in any 16 bit segment do not have to be written. If any 16 bit segment has fewer than four hexadecimal digits, it is assumed that the missing digits are leading zeroes.

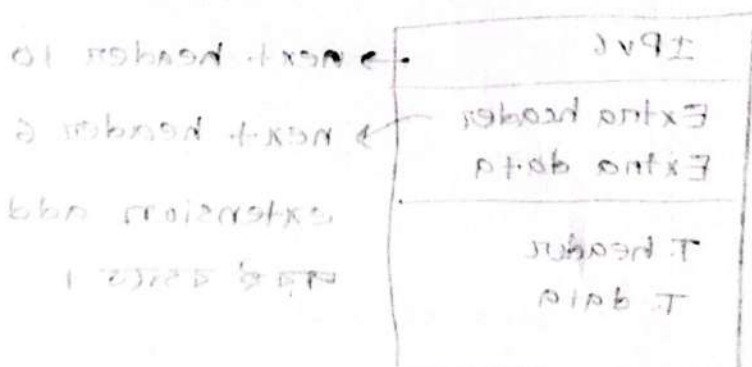
2031 : 0000 : 130F : 0000 : 0000 : 09C0 : 876A : 130B
 2031 : 0 : 130F : 0 : 0 : 9C0 : 876A : 130B

Rule 2: Any single, contiguous string of one or more 16 bit segments consisting of all zeroes can be represented once with a double colon '::', but can use it once.

1080 : 0 : 0 : 0 : 8 : 800 : 200C : 417A = 1080 :: 8 : 800 : 200C : 417A

extension → extra header / extra data

extension → extra header / extra data



TOPIC NAME : _____

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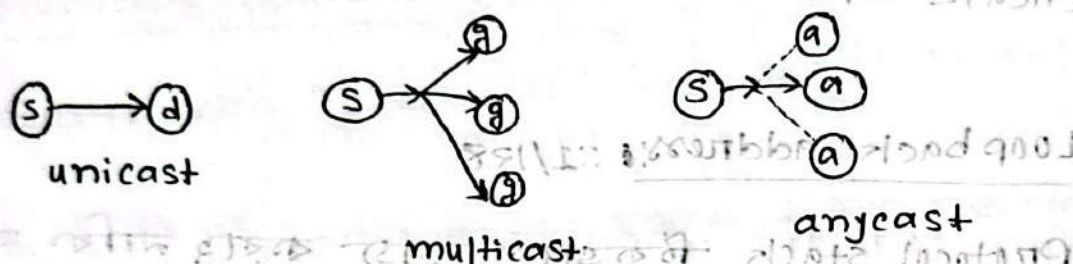
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Types of IPv6 addresses

Unicast : communicate specified 1 computer

multicast : communicate group of computers

Anycast : send group address that can receive multiple computers, but receive 1 computer

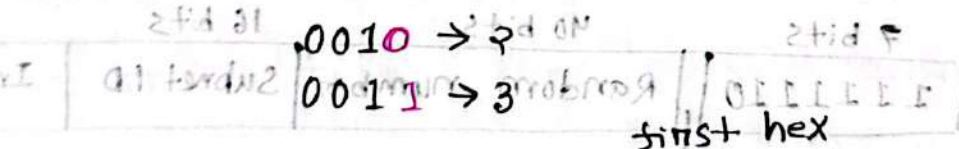


Unicast Global Address $2000::/3$

These are assigned by the IANA and used on public networks.

Typically they start at $2000::/3$

They fixed first 3 bits (001)



Unicast Global address value always will be 2 or 3



48 bit Mac address $\xrightarrow{EUI-64}$ 64 bit MAC

TOPIC NAME: _____

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Unspecified address: $:::/128$

কোনো software এর IP address প্রয়োজন, but IP address নেই, তখন use করা হয়। (128 bit fixed) এক্ষেত্রে শুধু নিজের সাথে communicate করা হয়। Internet বা অন্য কোনো device এর সাথে connection create করতে পারবে না।

Loopback address: $::1/128$

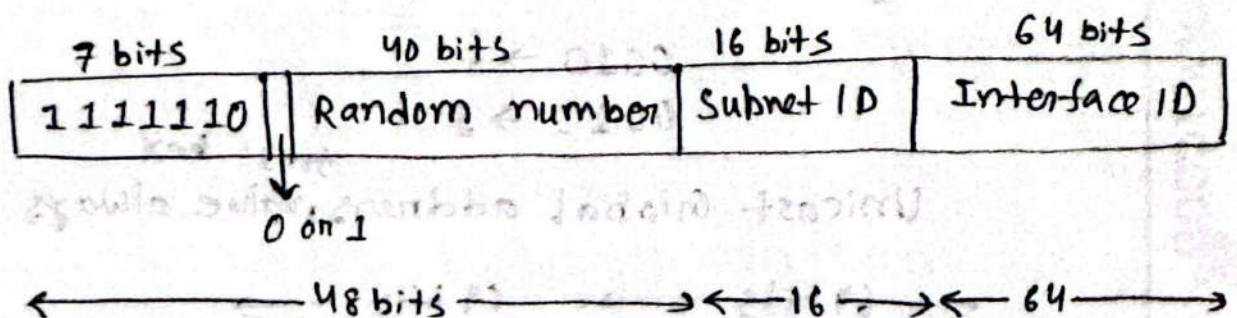
Protocol stack চিকিৎসা করা করছে না কি তা check করতে Loopback address দিায়।

127 bit $\rightarrow 0$, last 1 bit $\rightarrow 1$

Unique local Unicast address: $FC00::/7$

Globally unique

used in local communication



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Link Local Unicast Address: FE80::/10

These addresses refer only to a particular physical network

10 bits	38 bits	16 bits	64 bits
1111111010	All 0s	All 0s	Interface ID

Multicast Address: Consisting of all addresses that begin with "1111 1111" i.e "FF"

Anycast: group of device যা 1 জন receive করে।
একটি unicast address কে multiple device
এ set করে দিবে।

IPv4 to IPv6

Transition Techniques: 3 categories

i) Dual stack: Router both IPv4 and IPv6 run করে।

Router এর কাছে IPv4 packet আসে IPv4 network

দিয়ে পাঠিয়ে দেয়। IPv6 packet আসলে IPv6 net

দিয়ে পাঠায়।

Recommended option.

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ii) IPv6 Tunneling: protocol type 41.

IPv6 device আর IPv6 device to packet

প্যাকেট। Packet $\rightarrow$ IPv6 header + IPv6 data.

border gateway (dual stack) router $\rightarrow$ packet

আমরা পর it realise, IPv4 network দিয়ে

যাওয়া লাগবে। তাই- Router IPv4 header add

করে দিবে, যা IPv4 network দিয়ে যাবে।

port 41 দিয়ে পাঠানো হয়েছে। Next dual stack

router realise, this is an IPv4 header and

protocol type 41, so it identifies, এর আর

IPv6 এর data header আর। so it removes

IPv4 header and send it to the device.

iii) NAT Protocol Translation: IPv6 device যখন

IPv4 device এর সাথে connect করতে চায়,

তখন router IPv6 address change করে IPv4

বানায়। আরও IPv4 যদি IPv6 এর সাথে

connect করে (reply) তখন router IPv4 address

change করে IPv4 বানায়।

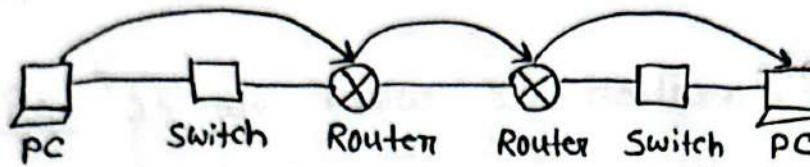
GOOD LUCK

TOPIC NAME: Data Link Layer

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Data link layer ensures hop to hop connection.



Nodes: hosts and routers

Links: wired / wireless

End device (PC) → Router

Router → Router

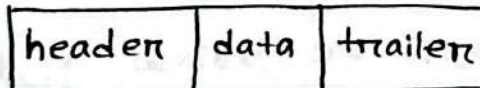
Router → End device (PC)

We ignore switch here.

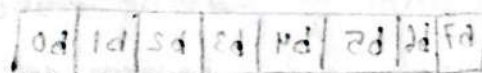
} hop

Communication between hops is a responsibility which the data link layer maintains.

Frame: layer-2 Packet



Use MAC Address



Link Layer Functions

- i) Framing
- ii) Link access
- iii) Reliable delivery between adjacent nodes

Why both link-level and end-to-end reliability?

Both are needed because link-level reliability improves performance by correcting local errors, while end-to-end

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reliability is essential to guarantee correct data delivery across the entire network. Neither alone is sufficient.

iv) error detection

v) error correction

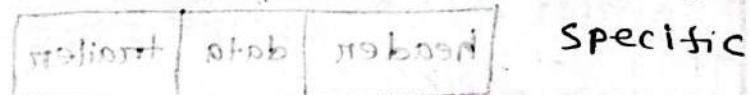
vi) flow control

vii) half duplex and full duplex

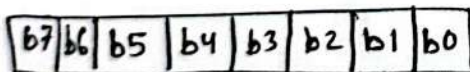
MAC Address: 12 hexadecimal digits (6 octets)

1st 6 digits → organizationally unique Identifier (OUI)

Last 6 digits → Network Interface Controller (NIC)



1st octet:



→ 0: Unicast

→ 1: Multicast

→ 0: globally unique

→ 1: locally administered

TOPIC NAME: _____

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Multicast MAC Address: 1st 6 hex digit : 01005E

Broadcast MAC Address: Dest MAC: FF-FF-FF-FF-FF-FF

ARP: If we know IP address, ARP will find out the MAC address.

ARP: Plug and Play

Dest MAC	Source MAC	Dest IP Add	Source IP	ARP Req/Reply
FF-FF-FF-FF-FF-FF	71-65-F7-2B-08-53	137.196.7.14	137.196.7.23	ARP Request
71-65-F7-2B-08-53	58-23-D7-FA-20-80	137.196.7.23	137.196.7.14	ARP Reply

Routing to another LAN

A and B device are connected to router R.

1st step: A → R (default gateway to MAC address known)
ARP request sent

2nd step: Router (knows destination address) is connected

(is directly connected. (same lane))

So we know the MAC address of B.

R → B

10 | 10

10 | 10

10 | 10

10 | 10

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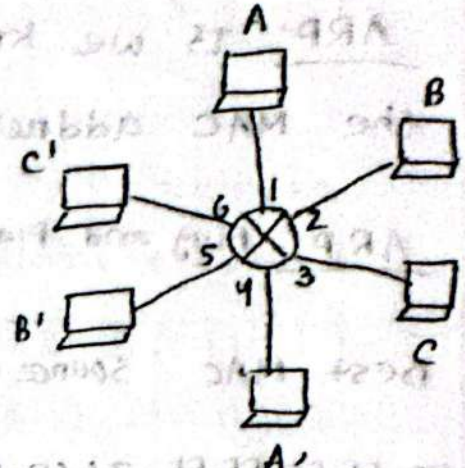
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Switch forwarding table

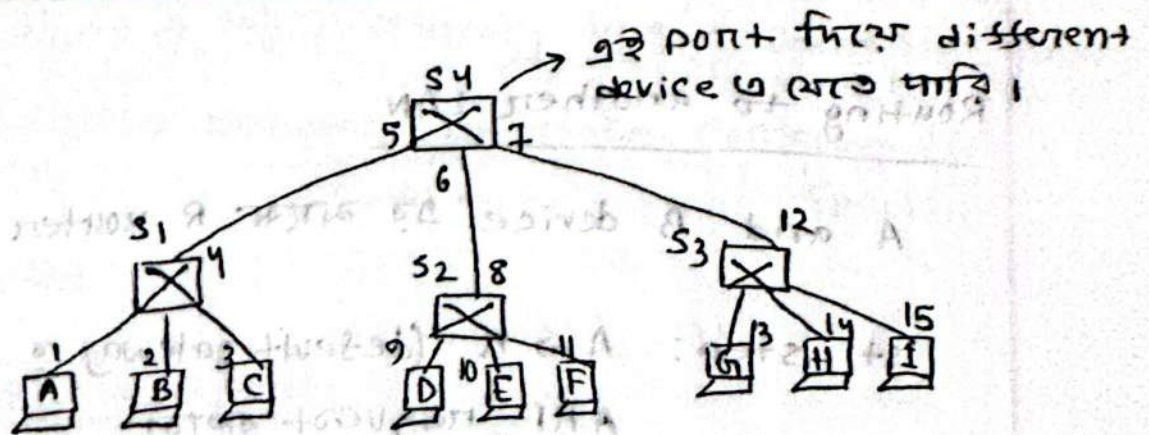
PC A wants to send something to PC A'.

MAC Address	Interface
A	1
A'	4



Selective forwarding

Interconnecting switches



Source destination

A → G
G → A

S1
A 1
G 4

S4
A 5
G 7

S2
A 8
G 10

S3
A 12
G 13

GOOD LUCK